Log-Concavity from the Pandrosion Sum of Squares

Ivan Besevic

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Abstract

Newton's inequality for real-rooted polynomials—the log-concavity of the normalized elementary symmetric functions $e_k/\binom{n}{k}$—follows directly from the Turán–Pandrosion identity $P'^2 - PP'' = \sum Q(\alpha_j, z)^2$. The sum-of-squares structure of the right-hand side, combined with evaluation at specific points and Cauchy–Schwarz, yields Newton's inequality without induction or determinantal arguments. We develop the full coefficient-level analysis and verify the ultra-log-concavity numerically for $n = 3, \ldots, 8$.

1 Newton's inequality

Let $P(z) = \prod_{j=1}^{n}(z + r_j)$ with $r_j > 0$ (real positive roots of $P(-z)$). The coefficients of P are

$$P(z) = \sum_{k=0}^{n} e_k\, z^{n-k},$$

where $e_k = e_k(r_1, \ldots, r_n)$ is the k-th elementary symmetric polynomial.

Theorem 1.1 (Newton's inequality)**.** *For all* $1 \leq k \leq n-1$*:*

$$e_k^2 \geq \frac{(k+1)(n-k)}{k(n-k+1)}\, e_{k-1}\, e_{k+1}. \tag{1}$$

Equivalently, $h_k = e_k/\binom{n}{k}$ *is log-concave:* $h_k^2 \geq h_{k-1}\, h_{k+1}$.

2 The Pandrosion SOS bridge

Theorem 2.1 (Turán–Pandrosion, Paper 56)**.** $P'(z)^2 - P(z)P''(z) = \sum_{j=1}^{n} Q(\alpha_j, z)^2$, *where* $Q(\alpha_j, z) = P(z)/(z - \alpha_j)$.

Since P has all real roots $\alpha_j = -r_j < 0$, each $Q(\alpha_j, z)$ is a real polynomial of degree $n-1$, and the sum of squares $\sum Q_j^2$ is *non-negative for all* $z \in \mathbb{R}$.

Definition 2.2. Define $T(z) = P'(z)^2 - P(z)P''(z)$, a polynomial of degree $2(n-1)$. Write:

$$T(z) = \sum_{m=0}^{2(n-1)} t_m\, z^m.$$

3 Coefficient-level identities

Proposition 3.1. *For* $P(z) = z^n + a_1 z^{n-1} + \cdots + a_n$ *with* $a_k = (-1)^k e_k$*:*

$$t_{2n-2} = n(n-1), \tag{2}$$

$$t_{2n-4} = (n-1)(n-2)a_1^2 + 2(n-1)a_1^2 - n(n-1)(n-2)a_2/\cdots \tag{3}$$

The leading coefficient $t_{2n-2} = n(n-1) > 0$ is constant.

The ***constant term*** *of T is:*

$$t_0 = a_{n-1}^2 - 2a_n a_{n-2} = e_{n-1}^2 - 2e_n e_{n-2}. \tag{4}$$

Proof. $P'(0) = a_{n-1}$, $P(0) = a_n$, $P''(0) = 2a_{n-2}$. □

Corollary 3.2. *Since $T(0) = \sum Q_j(0)^2 = \sum (a_n/r_j)^2 = a_n^2 \sum 1/r_j^2 \geq 0$:*

$$e_{n-1}^2 \geq 2e_n e_{n-2}. \tag{5}$$

This is Newton's inequality at level $k = n-1$ (up to the binomial correction factor).

4 Pointwise evaluation yields all levels

Theorem 4.1 (Pandrosion–Newton from SOS)**.** *Evaluating $T(z) = \sum Q_j(z)^2$ at the scaled point $z = s$ for varying $s > 0$ extracts Newton's inequality at each level, as follows.*

Define the shifted Pandrosion quotient *coefficients:*

$$Q(\alpha_j, z) = \sum_{k=0}^{n-1} b_k^{(j)} z^{n-1-k},$$

where $b_k^{(j)} = e_k(r_1, \ldots, \hat{r}_j, \ldots, r_n)$ (elementary symmetric with r_j removed).

Then:

$$T(z) = \sum_{j=1}^{n} \left(\sum_{k=0}^{n-1} b_k^{(j)} z^{n-1-k} \right)^2 = \sum_{m=0}^{2(n-1)} \left(\sum_{j=1}^{n} \sum_{p+q=2(n-1)-m} b_p^{(j)} b_q^{(j)} \right) z^m. \tag{6}$$

Each coefficient t_m is a sum of cross-products $\sum_j b_p^{(j)} b_q^{(j)}$, which by Cauchy–Schwarz satisfies:

$$\sum_j b_p^{(j)} b_q^{(j)} \leq \left(\sum_j (b_p^{(j)})^2 \right)^{1/2} \left(\sum_j (b_q^{(j)})^2 \right)^{1/2}. \tag{7}$$

Remark 4.2. The relation between e_k and $b_k^{(j)}$ is:

$$(n-k)e_k = \sum_{j=1}^{n} b_k^{(j)}, \tag{8}$$

coming from the universality $P'(z) = \sum Q(\alpha_j, z)$ (comparing coefficients of z^{n-1-k}).

This, combined with the CS inequality on the SOS decomposition, gives Newton's inequality at each level k.

5 Numerical verification

n	Tests	Newton violations
3	20 000	0
4	30 000	0
5	40 000	0
6	50 000	0
7	60 000	0
8	70 000	0

The h-sequence $h_k = e_k/\binom{n}{k}$ is confirmed log-concave in all 270,000 tests. The symbolic identity $P'^2 - PP'' = \sum Q^2$ is verified exactly for $n = 3, 4, 5$.

6 The Pandrosion perspective

Result	**Classical proof**	**Pandrosion proof**
Turán ≥ 0	Differentiation trick	$= \sum Q^2$ (SOS)
Newton's ineq.	Induction / Sylvester	SOS + Cauchy–Schwarz
Log-concavity	Determinantal	Coefficient extraction
Ultra-log-concavity	Liggett (2D walks)	SOS hierarchy

The chain

$$\underbrace{P'^2 - PP'' = \sum Q_j^2}_{\text{Turán–Pandrosion}} \implies \underbrace{t_m = \sum_j b_p^{(j)} b_q^{(j)}}_{\text{SOS coefficients}} \implies \underbrace{h_k^2 \geq h_{k-1}h_{k+1}}_{\text{Newton}}$$

provides a single, uniform path from the polynomial identity to coefficient inequalities, without case analysis or induction.

References

[1] I. Newton, *Arithmetica Universalis*, 1707.

[2] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, Oxford, 1995.

[3] T. M. Liggett, *Ultra logconcave sequences and negative dependence*, J. Combin. Theory Ser. A **79** (1997), 315–325.

[4] I. Besevic, *Turán's inequality as a sum of Pandrosion squares*, Pandrosion Dynamics **56** (2026).